

# ECE250: Signals & Systems Quiz-1

**Date:** 27-Aug-2025

**Total Points:** 10

**Duration:** 15 Minutes

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**Note:**

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
  2. Institute Plagiarism policy are strictly applicable.
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**[CO1] [Q1] [5 Points]:** Determine whether or not the below signals are periodic. If periodic, determine the fundamental time period.

(a)

$$x(t) = e^{j(\pi t - 1)}$$

(b)

$$x[n] = \cos\left(\frac{\pi n^2}{8}\right)$$

**[CO1] [Q2] [5 Points]:** Consider the below discrete-time signal:

$$x[n] = -1 + \sum_{k=2}^{\infty} \delta[n - k]$$

(a) Plot this signal.

(b) Determine the values of constants  $A$ ,  $M$  and  $n_0$ , so that  $x[n]$  may be expressed as  $x[n] = Au[Mn + n_0]$ .